Product summary

UBX-F11 series

u-blox F11 standard precision GNSS chips

Standard

Professional

Automotive

Ultra-low-power GNSS with seamless L1/L5 dual band use for best accuracy in all environments

- 7 mW in Low Energy Accurate Positioning (LEAP) mode
- Selective use of L5 band for best balance of power/accuracy
- L1 or L1/L5 band support with a single chip
- Application specific modes for easy implementation
- Upgradable firmware and embedded flash

UBX-F11170-CC

2.39 x 2.39 x 0.36 mm



UBX-F11070-KB
UBX-F11270-KB

5.0 x 5.0 x 0.85 mm



Product description

The u-blox F11 GNSS receiver provides exceptional meter-level positioning accuracy in all environments. The UBX-F11 chips selectively take advantage of L1/L5 dual-band operation to provide the best balance of power and position accuracy.

Depending on the environment, the Low Energy Accurate Positioning (LEAP) mode reduces the power consumption to 7 mW while retaining superior position accuracy, which enables great power autonomy for all battery-operated devices. Modes for applications such as vehicle tracking, swimming, and hiking allow easy firmware implementation and optimized performance.

The free AssistNow Predictive Orbits data download facilitates TTFF in <2 s. The Live Orbits data download furthermore improves the position accuracy after cold starts.

The UBX-F11 chips, available in two different packages, support selective L1/L5 band operation. All chips are pin-compatible with the previous u-blox 10 generation.

The u-blox F11 family detects jamming and spoofing events and reports them to the host, so that the system can react to such events. Advanced filtering algorithms mitigate the impact of RF interference and jamming, thus enabling the product to operate as intended.

u-blox F11 chips are qualified according to JEDEC JESD47 and are manufactured in IATF 16949 certified sites.

	UBX-F11070-KB	UBX-F11170-CC	UBX-F11270-KB
Grade			
Automotive			
Professional	•		•
Standard		•	
GNSS			
GPS + QZSS/SBAS	•	•	•
Galileo	•	•	•
BeiDou	•	•	•
NavIC	•	•	•
Bands	L1/L5/ E5a/B2a	L1/L5/ E5a/B2a	L1/L5/ E5a/B2a
Interfaces			
UART	1	1	1
SPI	1	1	
DDC (I2C compliant)	1	1	
Features			
Embedded flash memory			•
Firmware upgradable		•	•
AssistNow lifetime access	P	P	P
RTC crystal	S	S	S
Oscillator	C/T	C/T	C/T
Antenna supervisor	S	S	S
Timepulse	1	1	1

L = Live orbits / P = Predictive orbits
S = supported, may require ext. components
C/T = Crystal and TCXO supported

UBX-F11 series

Product performance

Receiver type	u-blox F11 engine	
Bands	GPS	L1 C/A, L5
	BDS	B1I, B1C, B2a
	QZSS	L1 C/A, L1S, L1Sb, L5
	GAL	E1B/C, E5a
	SBAS	L1 C/A
	BDSBAS	B1C
	NavIC	L5
Nav. update rate	Up to 25 Hz (single GNSS) Up to 10 Hz (3 concurrent GNSS)	
Horizontal position accuracy [1]	1.0 m / 1.5 m CEP (with SBAS) 1.5 m / 2.0 m CEP (without SBAS)	
Acquisition [1]	Cold start	28 s
	Aided start	2 s
	Hot start	2 s
Sensitivity [1]	Tracking & Nav.	-167 dBm
	Reacquisition	-160 dBm
	Cold start	-148 dBm
	Hot start	-159 dBm

External components

Oscillator	TCXO
RTC input (optional)	32.768 kHz
Antenna supply & supervisor	External circuit required for short and open circuit detection
Flash memory (optional)	SPI interface

Tracking features

Odometer	Measures traveled distance with >99% accuracy
Data logging	Stores position/velocity/time information on flash memory without host interaction
Data batching	Stores position/velocity/time information over up to 10 min in RAM without host interaction
LEAP	Low energy accurate positioning reduces the power consumption while retaining superior position accuracy
Selective L1/L5 use	Smart use of L1/L5 dual band only when beneficial to improve the position accuracy

Security features

Signal integrity	RF interference & jamming detection and reporting Spoofing detection and reporting
Device integrity	Receiver configuration lock by command
Secure interface	Signed UBX messages (SHA-256) JTAG debug interface disabled by default

Electrical data

Power save mode		LEAP	L1	L1/L5
Power consumption [2]	2 GNSS	6 mW	12 mW	21 mW
	3 GNSS	7 mW	14 mW	25 mW
Supply voltage	0.9 V to 1.8 V			
Digital I/O voltage	1.8 V or 3.3 V			
Backup supply	1.65 V to 3.6 V			

Package

33 pin WL-CSP: 2.39 x 2.39 x 0.36 mm
28 pin QFN: 5.0 x 5.0 x 0.85 mm

Environmental data, quality & reliability

Operating temp.	–40°C to +85°C
Storage temp.	–40 °C to +125 °C
MSL	1
Environmental grade	2015/863/EU RoHS-3, Green, IEC-61249-2-21 halogen-free
Environmental testing	JEDEC JESD47
Quality management	Manufactured and fully tested in IATF 16949 certified production sites

Interfaces

Serial interfaces	1 UART 1 DDC (I2C compliant) 1 SPI
Digital I/O	1 configurable time pulse 1 EXTINT interrupt input
Raw data output	Code phase data, carrier phase data
Memory	SPI interface for optional Flash
Protocols	NMEA 4.11, UBX binary, RTCM 3.4

Compatible u-blox location services

AssistNow	Achieves premium performance in challenging IoT environments
CloudTrack	Ultra-low-power asset tracking service

Support products

EVK-F112	u-blox F11 GNSS evaluation kit with a firmware-upgradable chip and TCXO
u-center 2	Highly intuitive software for GNSS performance evaluation

Product variants

UBX-F11070-KB	u-blox F11 GNSS chip, 28 pin QFN GPS/GAL/BDS as default configuration
UBX-F11170-CC	u-blox F11 GNSS chip, 33 pin WL-CSP, GPS/GAL/BDS as default configuration, upgradable firmware
UBX-F11270-KB	u-blox F11 GNSS chip, 28 pin QFN, GPS/GAL/BDS as default configuration, upgradable firmware, embedded flash

1 = GPS/Galileo/Beidou + SBAS/QZSS
2 = Supply voltage 0.9 V, digital I/O voltage 1.8 V, 1Hz navigation update rate

Further information

For contact information, see www.u-blox.com/contact-us.

For more product details and ordering information, see the product data sheet.

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